Venkata Lakshmi Parimala Pasupuleti

Senior Software Engineer · Python · Full-Stack · PropTech · Leasing Domain · AWS · AI-Assisted Development · Microservices

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McLean area, Virginia — Remote

PROFESSIONAL SUMMARY

Senior software engineer with 8 years of full-stack SaaS delivery for property managers — the exact persona AppFolio's Leasing team builds for. Built and owned **tenant application workflows, lease processing, move-in/move-out automation, and owner-facing financial tooling** at Entrata (a direct AppFolio competitor) across **20+ modules, 1M+ records, 100K+ MAU, 99.5% uptime**. Leasel demonstrates this directly: an agentic RAG pipeline achieving **93.6% precision over 1,240 real lease PDFs** with a deterministic rules engine and CI/CD eval gates — same leasing domain this team builds for. On AI-assisted development: I built **AgentProbe** to evaluate AI agents against 5 quality metrics and gate failing runs in CI/CD — I don't just use AI tools, I build infrastructure to trust them in production. Led Docker microservices rollout (50% speedup), AWS S3/RDS/Glue migration (100% PHI compliance), first production AI pipeline on 1.6M financial records; mentored 5+ engineers, 300+ code reviews. M.S. Data Analytics Engineering, GMU (GPA 3.87, May 2026).

WORK EXPERIENCE

George Mason University
Aug 2024 – Aug 2025

Research Associate — ML Pipelines · Anomaly Detection · Statistical Modeling

- Built end-to-end ML pipelines (Python, R, 190+ countries) achieving **$R^2 = 0.977$**; applied Isolation Forest anomaly detection across 3 research pipelines; automated statistical reporting driving **40% improvement** in stakeholder comprehension.

Entrata
Jan 2021 – Aug 2024

Senior Software Engineer — Leasing & Owner Workflows · Full-Stack · AWS · Production AI · Mentorship

- Owned **leasing, tenant, and owner-facing workflows** end-to-end — tenant applications, lease processing, move-in/move-out automation, distribution reporting, and bulk workflow operations across **20+ modules** (1M+ records, 100K+ MAU, 99.5% uptime, 120+ Agile sprints) for the same property managers AppFolio's Leasing team serves.
- Led **three platform-level initiatives**: Docker microservices rollout (50% execution speedup), AWS S3/RDS/Glue migration (100% PHI compliance, zero incidents), and first production AI pipeline on **1.6M real financial records** (PyTorch, OpenAI API) with quality gates and HITL review before customer exposure; optimized 200+ SQL queries (30% performance gain).
- Mentored **5+ engineers**, conducted **300+ peer code reviews**; championed AI-assisted development practices and built open-source evaluation infrastructure (AgentProbe) to gate AI outputs in CI/CD — bringing engineering rigor to LLM integration.

Entrata
Jan 2019 – Dec 2020

Software Engineer — Compliance Automation · API Integrations · Data Pipelines

- Built HUD/LIHTC compliance modules with deterministic rules engine eliminating **80% of manual workload**, zero compliance violations across 100K+ units; built API integrations and webhooks across property management systems; 300+ peer code reviews.

Entrata
Jan 2017 – Dec 2018

Associate Engineer — Full-Stack · Python · PHP · PostgreSQL · REST APIs

- Built and shipped full-stack features and SQL data layers (PHP, JavaScript, PostgreSQL) for leasing and property management workflows; 0→1 prototyping through production rollout across 100K+ unit portfolios.

PROJECTS

Leasel: Agentic Compliance Pipeline for Legal Documents
PROPTECH AGENTIC RAG

Python · LangGraph · ChromaDB · Deterministic Rules Engine · LLM-as-Judge — github.com/venkatalakshmiparimala/Lease_complaince_rag

- **93.6% precision on 1,240 lease PDFs** — the same leasing domain this team builds for; agentic LangGraph RAG pipeline with deterministic rules engine for fully auditable outputs; pass@k eval harness (pass@1 81.7%, pass@3 88.3%) with LLM-as-judge scoring and CI/CD eval gates.

AgentProbe: Multi-Agent AI Eval Framework
MULTI-AGENT LLM-AS-JUDGE

Python · LangGraph · FastAPI · ChromaDB · React · Docker · GitHub Actions — github.com/venkatalakshmiparimala/AgentProbe

- 3-agent LangGraph runtime with tool calling, inter-agent state handoffs, **5-metric LLM-as-judge eval engine**, CI/CD regression gates blocking failing agent runs before production, 15-endpoint FastAPI backend, and **React observability dashboard** — AI-assisted coding infrastructure, not just AI-assisted coding.

AI on FHIR: HIPAA-Compliant Medical Records NLP Platform
HEALTHCARE

Python · Flask · React · FHIR APIs · BERT · LLMs · Docker — github.com/venkatalakshmiparimala/AI_on_FHIR

- ~60% reduction in clinical retrieval time — NLP platform translating natural-language queries into FHIR API calls via LLMs + BERT; HIPAA-compliant RBAC, end-to-end from design to deployment.

EDUCATION

M.S. Data Analytics Engineering — George Mason University — May 2026 · GPA 3.87

ML systems, statistical modeling, data engineering, AI pipelines, system design, data governance

B.Tech Computer Science & Engineering — JNTU Anantapur — 2013 – 2017

TECHNICAL SKILLS

Languages & Backend: Python, JavaScript, TypeScript, PHP, SQL, PL/SQL, R; FastAPI, Django, Flask, REST APIs, GraphQL (familiar), webhooks, API integrations, RabbitMQ, SOA, event-driven distributed systems; Ruby (fast ramp — Nice-to-Have in JD, OOP/dynamic-typed foundation)

Frontend & Data: React, Redux (familiar), HTML/CSS; PostgreSQL, MySQL, query optimization, data pipelines, ETL, AWS Glue; LangGraph, LangChain, RAG Pipelines, LLM-as-Judge, OpenAI API, Anthropic API, PyTorch, BERT

Infrastructure & Cloud: AWS (S3, EC2, RDS, Glue), Docker, Microservices, CI/CD, GitHub Actions, test-driven development, Agile/Scrum, Kanban, HIPAA, PHI compliance, scalable architecture, Git, Linux